

Yogitha A S

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CAREER OBJECTIVE

Computer Science student with a strong foundation in data structures, databases, and backend engineering, and hands-on experience in container security and applied AI. Seeking to apply my skills in a challenging role in software development.

EDUCATION

B.Tech in Computer Science and Engineering, PES University, Bengaluru — Pursuing (2023 – Present)

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript

CS Fundamentals: Data Structures and Algorithms, Operating Systems, Computer Networking, DBMS, Compiler Design, OOP

AI / ML: Generative AI, RAG, Machine Learning

DevOps and Tools: Docker, Kubernetes, GitHub, Office365

Backend and Web: Node.js, React.js, Spring Boot, REST APIs, HTML, CSS, MongoDB

Soft Skills: Problem Solving, Team Collaboration, Adaptability, Communication

RESEARCH EXPERIENCE

Research Intern

2024 – 2025

ISFCR SOC Lab, PES University

- Researched container security using Docker and Kubernetes; built an end-to-end DoS attack detection and prevention tool in Python.
- Integrated Falco, Wazuh, Snort, and OSSEC for log analysis and real-time threat monitoring; documented Kubernetes security mitigation strategies in DevOps workflows.
- Co-authored and published an IEEE conference paper on Kubernetes runtime security using Falco, VirusTotal, and Talon.

ACADEMIC PROJECTS

AI-Powered Placement and Career Assistant (Ongoing)

Python, NLP, Machine Learning, Generative AI

- Built a capstone platform integrating semantic resume-job matching using ColBERT and ConFit v2 with ChromaDB for ATS-friendly job-candidate recommendations.
- Developed an adaptive aptitude assessment engine using Bayesian Knowledge Tracing (BKT), EMA smoothing, and an epsilon-greedy strategy to dynamically adjust question difficulty per candidate.
- Built an AI-powered mock interview system with real-time feedback using MobileNetV2, ViT-B/16, and Whisper + Ollama for behavioral analysis, speech-to-text, and response evaluation, plus skill-gap analysis and career-path recommendations.

IaC / Kubernetes Manifest Risk Reviewer

Python, LangGraph, RAG, Kubernetes, FastAPI

- Built an agentic GenAI system that reviews Terraform/Kubernetes YAML changes on live GitHub PRs, grounded in CIS benchmarks, with low/medium/high risk-tiered auto-fix, approval, and escalation routing.
- Combined hybrid BM25 + semantic retrieval with deterministic policy checks (wildcard RBAC, privileged containers) and kube-linter validation to reduce false confidence in LLM output.
- Implemented a persistent human-approval workflow (LangGraph interrupt/resume + SQLite checkpointer) with a live GitHub webhook (FastAPI + ngrok) posting real-time AI risk-review comments on PRs.

Supply Chain Resilience and Demand Sensing Agent

Python, LLaMA-3-8B, RAG, LangChain

- Built an agentic AI system for pharmaceutical procurement using hybrid semantic search (RAG + cross-encoder reranking) and LLaMA-3-8B, with a dashboard integrating a real-time news monitor, inventory tracker, and compliance verifier.

Self Help Groups Financial Tracker

Java, Spring Boot, MySQL, Thymeleaf

- Built a full-stack web app managing transactions, savings, loans, and advisory services with role-based access control and monthly financial reporting.
- Designed with Spring MVC, Spring Data JPA, and MySQL/H2, following MVC architecture; developed using Maven and JUnit 5.

CERTIFICATIONS

- IEEE Conference Publication — Real-Time Container Threat Detection and Response Using Falco, VirusTotal and Talon.
- 30 Day Python using AI — AI For Techies.
- AI Tools and Claude Workshop — be10x.
- JoEL Roadshow 2024, PES University — presented a Bluetooth-controlled Disaster Retrieval Robotic Arm Car (ESP32 + Arduino).
- Infosys Springboard certifications: OpenAI GPT-3 for Developers, Prompt Engineering, Introduction to OpenAI GPT Models.
- CIE IGNITE Season 01 (Stage 2 Team Pitch) — pitched “Asha,” an AI-powered legal rights platform for women.

EXTRACURRICULAR ACTIVITIES

- Hospitality Head, GRONIT Club (Green Computing Club) — organised hackathons and workshops for sustainability/green tech initiatives.
- Member, Shunya (Mathematics Club) — participated in academic events and problem-solving sessions.